Comprehensive Air Quality Assessment Report
Hay Hassani Monitoring Station, Casablanca - 2014

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Executive Summary

This report presents a comprehensive analysis of air quality data from the Hay Hassani monitoring station in Casablanca for the calendar year 2014. The assessment reveals significant insights into pollution patterns, meteorological influences, and regulatory compliance status according to Moroccan environmental standards (Decree 2-09-286 of December 8, 2009). Primary pollutants such as NO, SO, and CO comply with standards, while PM and O exhibit persistent exceedances, requiring targeted corrective measures.

1 Introduction

1.1 Study Framework

The Hay Hassani monitoring station is an urban background site in Casablanca, providing critical data on atmospheric pollution levels. This analysis covers the full 2014 dataset, comprising 8,760 hourly measurements of multiple pollutants (NO, O, PM, SO, CO) and meteorological parameters (wind direction and speed, temperature, humidity). The analyzed period spans from January 1, 2014, at 00:00 to January 1, 2015, at 00:00, covering 366 days.

1.2 Data Coverage and Quality

The dataset demonstrates excellent completeness, with coverage exceeding 87% for all measured parameters. Data quality is rated as good to excellent across all pollutants.

Table 1: Data coverage by pollutant

Pollutant	**Coverage (%)**	**Evaluation**
NO_2	99.0	Excellent
O_3	94.7	Good
PM_{10}	87.2	Good
SO_2	96.3	Excellent
CO	93.8	Good

2 Regulatory Compliance

Standards are established by Decree No. 2-09-286 of December 8, 2009, which defines limit values for the protection of human health and vegetation.

2.1 Overall Compliance Status

The station demonstrates mixed compliance with Moroccan air quality standards. Primary gaseous pollutants are compliant, but particulate matter and ozone show non-compliances.

Table 2: Regulatory compliance status

Pollutant	Status	Observations
NO_2	COMPLIANT	Meets all limits
SO_2	COMPLIANT	Levels well below limits
PM_{10}	NON-COMPLIANT	Exceeds 90.4th percentile daily
O_3	NON-COMPLIANT	Exceeds 8-hour averages and consecutive days
CO	COMPLIANT	Levels below limits

2.2 Detailed Compliance Results

Table 3: Detailed compliance results

Pollutant	Parameter	Value	Limit	Status
NO_2	Annual mean	$18.9\,\mu g\,m^{-3}$	$50\,\mu g\,m^{-3}$	COMPLIANT
	98th percentile hourly	$71.2\,\mu g\,m^{-3}$	$200\,\mu g\,m^{-3}$	COMPLIANT
SO_2	Annual mean	$4.0\,\mu g\,m^{-3}$	$20\,\mu g\,m^{-3}$	COMPLIANT
	99.2th percentile daily	$17.5\,\mu g\,m^{-3}$	$125\,\mu g\,m^{-3}$	COMPLIANT
PM_{10}	90.4th percentile daily	$62.7\,\mu g\,m^{-3}$	$50\,\mu g\,m^{-3}$	NON-COMPLIANT
CO	Maximum daily 8h	$2.68\,mg\,m^{-3}$	$10\,mg\,m^{-3}$	COMPLIANT
O_3	Maximum 8h	$110.2\,\mu g\,m^{-3}$	$110\,\mu g\,m^{-3}$	NON-COMPLIANT
	Consecutive days >65	12 days	3 days	NON-COMPLIANT

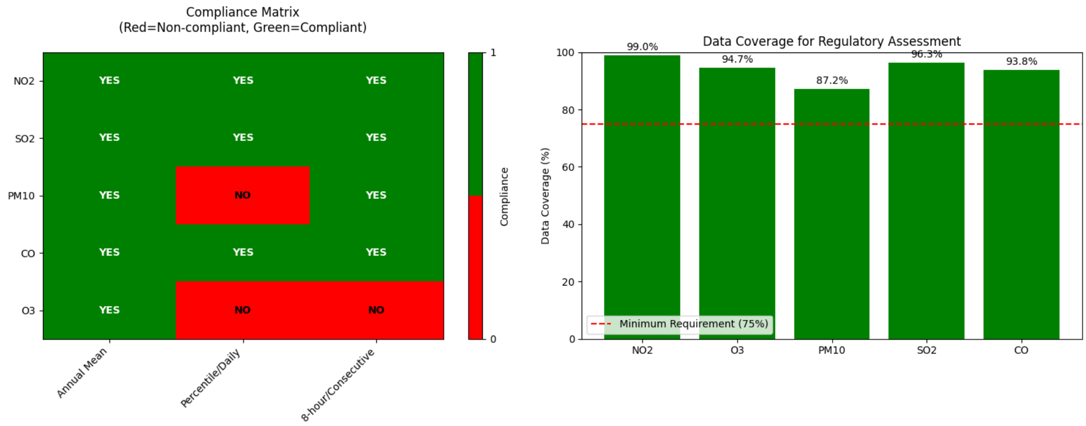


Figure 1: Compliance matrix (Red: Non-compliant, Green: Compliant) and data coverage for regulatory assessment.

Percentile calculations use the linear interpolation method recommended by WHO, with a minimum 75% data completeness requirement, which is met for all pollutants.

3 Pollutant Concentration Analysis

3.1 Descriptive Statistics

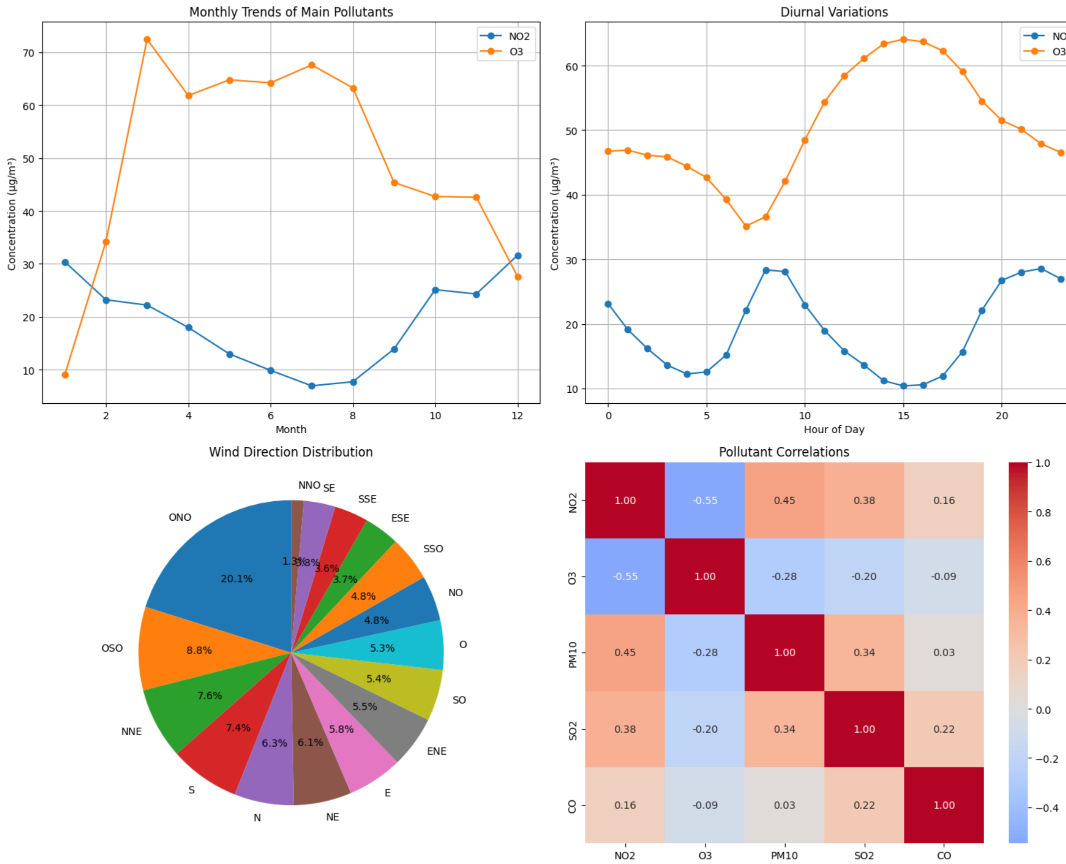
Table 4: Annual pollutant statistics

Pollutant	Mean	Maximum	95th Percentile	Exceedances
NO_2	18.9 µg m^{-3}	157.7 µg m^{-3}	55.3 µg m^{-3}	1,064 hours (>40)
O_3	50.6 µg m^{-3}	149.6 µg m^{-3}	93.6 µg m^{-3}	140 hours (>100)
PM_{10}	33.8 µg m^{-3}	324.1 µg m^{-3}	76.5 µg m^{-3}	1,612 hours (>45)
SO_2	4.0 µg m^{-3}	83.1 µg m^{-3}	11.2 µg m^{-3}	123 hours (>20)
CO	0.787 mg m^{-3}	3.894 mg m^{-3}	1.542 mg m^{-3}	0 hours (>10)

Annual mean concentrations are relatively low for gaseous pollutants, but maximum values indicate episodic high pollution, likely linked to local sources or adverse meteorological conditions.

3.2 Temporal Trends

Monthly trends show clear seasonal variations: NO and PM are higher in winter and autumn, while O peaks in spring and summer due to photochemical formation.



	NO2	O3	PM10	SO2	CO
NO2	1.00	-0.55	0.45	0.38	0.16
O3	-0.55	1.00	-0.28	-0.20	-0.09
PM10	0.45	-0.28	1.00	0.34	0.03
SO2	0.38	-0.20	0.34	1.00	0.22
CO	0.16	-0.09	0.03	0.22	1.00

Figure 2: Monthly trends of key pollutants (NO2, O3), diurnal variations (NO2, O3), wind direction distribution, and pollutant correlations.

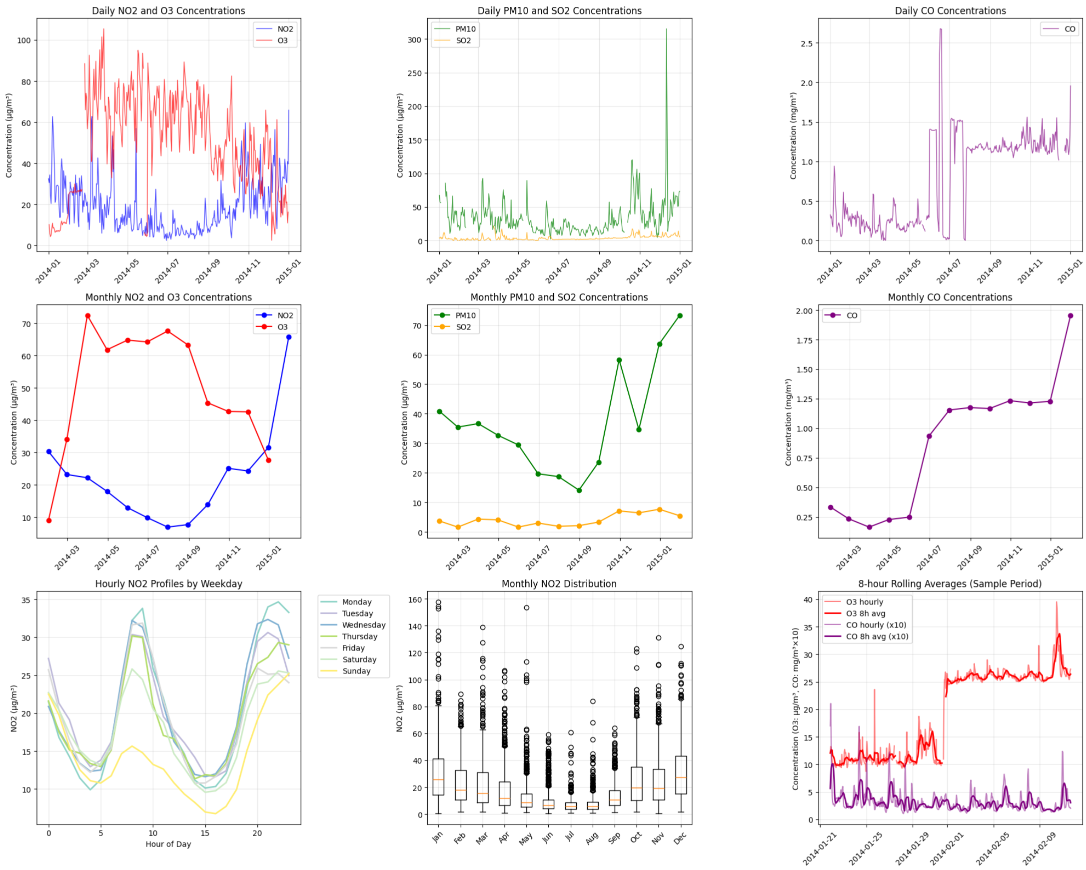


Figure 3: Daily concentrations of NO2 and O3, PM10 and SO2, CO; monthly concentrations of NO2 and O3, PM10 and SO2, CO; hourly NO2 profiles by weekday; monthly NO2 distribution; 8-hour rolling averages (sample period).

Diurnal profiles reveal morning and evening peaks for NO, corresponding to traffic rush hours, with a maximum weekday-weekend difference of 11.8 g/m^3 at 9:00 AM.

3.3 Seasonal and Diurnal Profiles

Table 5: Seasonal concentration variations

Pollutant	Winter	Spring	Summer	Autumn
NO_2	High (28.1)	Moderate (17.7)	Low (8.9)	Moderate (27.0)
O_3	Low (23.6)	High (66.4)	High (65.2)	Moderate (43.6)
PM_{10}	High (46.7)	Moderate (32.9)	Low (17.5)	High (42.2)
SO_2	Moderate (4.3)	Moderate (4.1)	Low (3.2)	Moderate (4.5)
CO	High (0.92)	Moderate (0.75)	Low (0.65)	Moderate (0.88)

Seasonal variations indicate meteorological influence: winter thermal inversions favor primary pollutant accumulation, while high summer solar radiation promotes ozone formation.

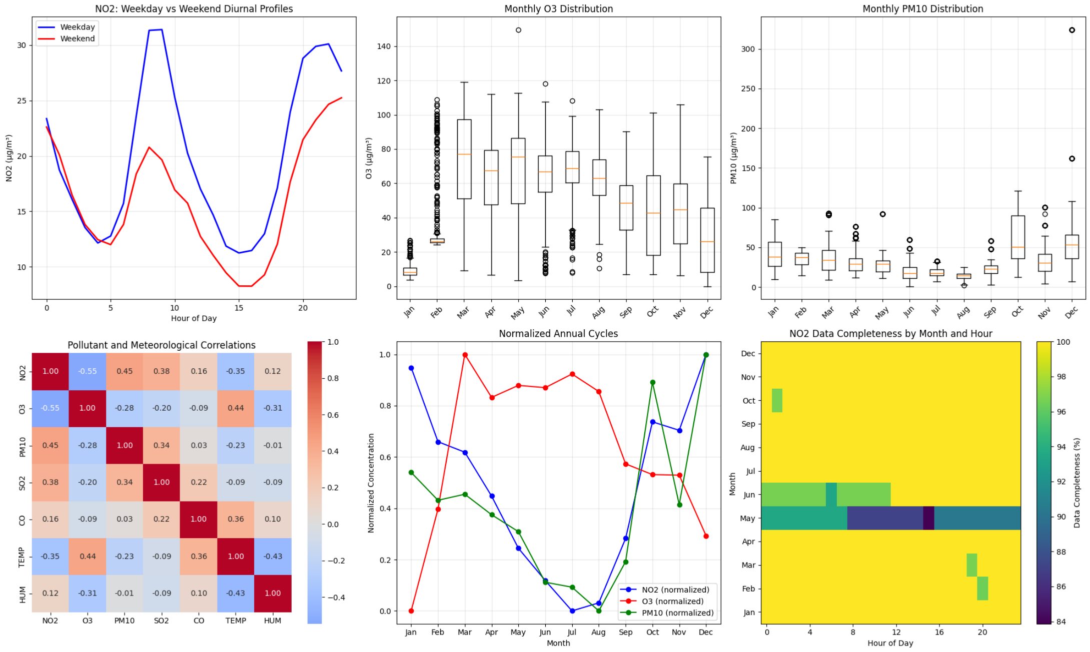


Figure 4: Diurnal NO2 profiles (weekday vs weekend), monthly O3 distribution, monthly PM10 distribution, pollutant-meteorology correlations, normalized annual cycles, NO2 data completeness by month and hour.

4 Meteorological Analysis

4.1 Climatic Parameters

Table 6: Annual meteorological parameters

Parameter	Value
Mean temperature	16.9 °C
Annual thermal range	26.2 °C
Daily thermal amplitude	4.9 °C
Mean relative humidity	77.1 %
Mean wind speed	$2.16\,\mathrm{m\,s^{-1}}$
Dominant wind direction	WNW (19.7 %)

Temperature ranges from 5.2°C to 31.5°C, with an average diurnal amplitude of 4.9°C. Humidity is high, averaging 77.1%, potentially favoring secondary particle formation.

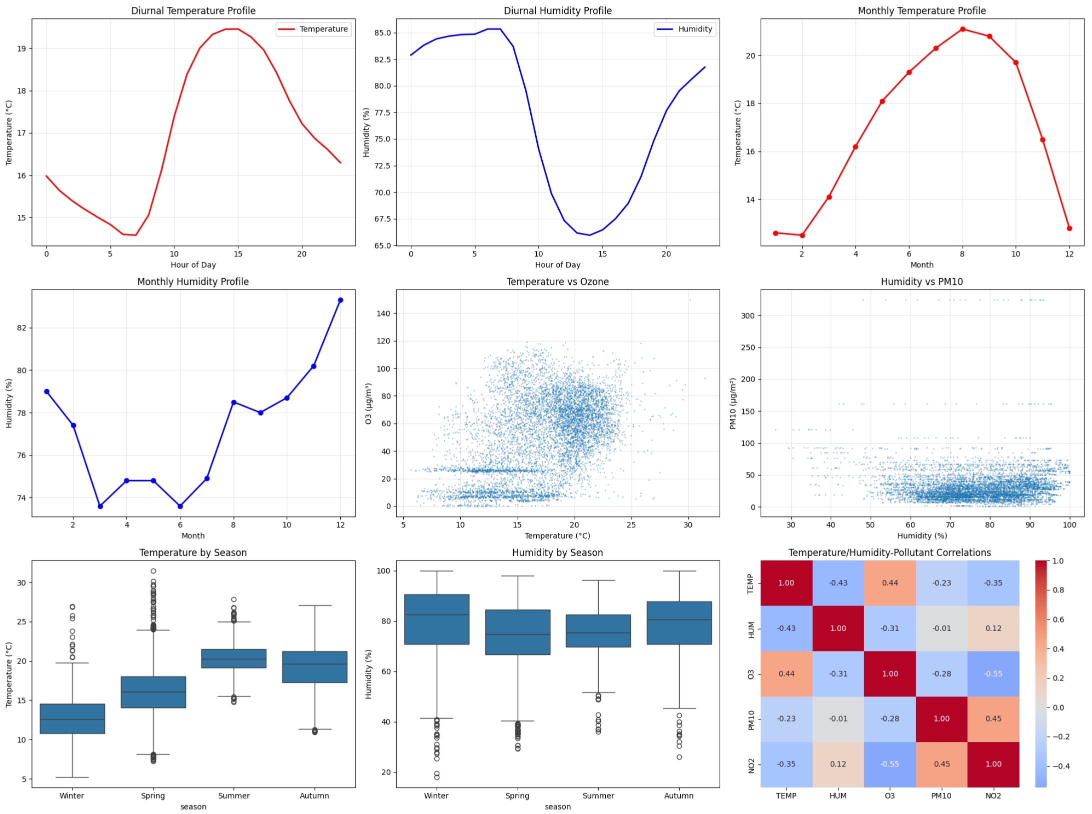


Figure 5: Diurnal temperature and humidity profiles, monthly temperature and humidity profiles, temperature vs ozone, humidity vs PM10, seasonal temperature, seasonal humidity, temperature/humidity/pollutant correlations.

4.2 Pollution-Meteorology Relationships

Table 7: Pollution-temperature-humidity correlations

Pollutant	Temperature Correlation	Humidity Correlation
O_3	+0.440 (strong)	-0.314 (moderate)
PM_{10}	-0.232 (weak)	-0.015 (none)
NO_2	-0.353 (moderate)	+0.123 (weak)
SO_2	-0.093 (none)	-0.094 (none)
CO	+0.357 (moderate)	+0.100 (weak)

Ozone shows a strong positive correlation with temperature, indicating increased photochemical production in summer. Nocturnal inversions (average strength 1.8°C) increase NO concentrations by 17% at night compared to daytime.

5 Wind Rose Analysis

5.1 Source Identification

Table 8: Wind sectors with high pollution

Pollutant	Dominant Sector	Concentration
NO_2	North (N)	$33.79\,\mu g\,m^{-3}$
O_3	South (S)	$49.12\,\mu g\,m^{-3}$
PM_{10}	Northeast (NE)	$47.35\,\mu g\,m^{-3}$
SO_2	Northeast (NE)	$6.14\,\mu g\,m^{-3}$
CO	North (N)	$0.90\,mg\,m^{-3}$

The detailed 16-sector analysis reveals distinct patterns:

- **NO_2 and CO**: Maximum concentrations from the North, characteristic of road traffic sources
- **PM_{10} and SO_2**: Peak from the Northeast, indicating industrial or combustion sources
- **O_3**: Maximum from the South, typical of photochemical formation and regional transport

5.2 Wind Speed Influence

Table 9: Impact of wind speed on pollution

Pollutant	Calm <1 m/s	Light	Moderate	Strong >5 m/s
NO_2	38.87	22.85	13.57	10.27
O_3	26.62	41.66	52.21	43.87
PM_{10}	46.77	40.59	34.79	27.56
SO_2	4.94	4.83	4.24	3.85
CO	1.01	0.69	0.57	0.70

Calm winds (<1 m/s) significantly amplify concentrations of NO (38.87 g/m^3), PM (46.77 g/m^3), and CO (1.01 mg/m^3), limiting pollutant dispersion. Ozone shows the opposite behavior, peaking at moderate wind speeds (3–5 m/s), favoring photochemical formation.

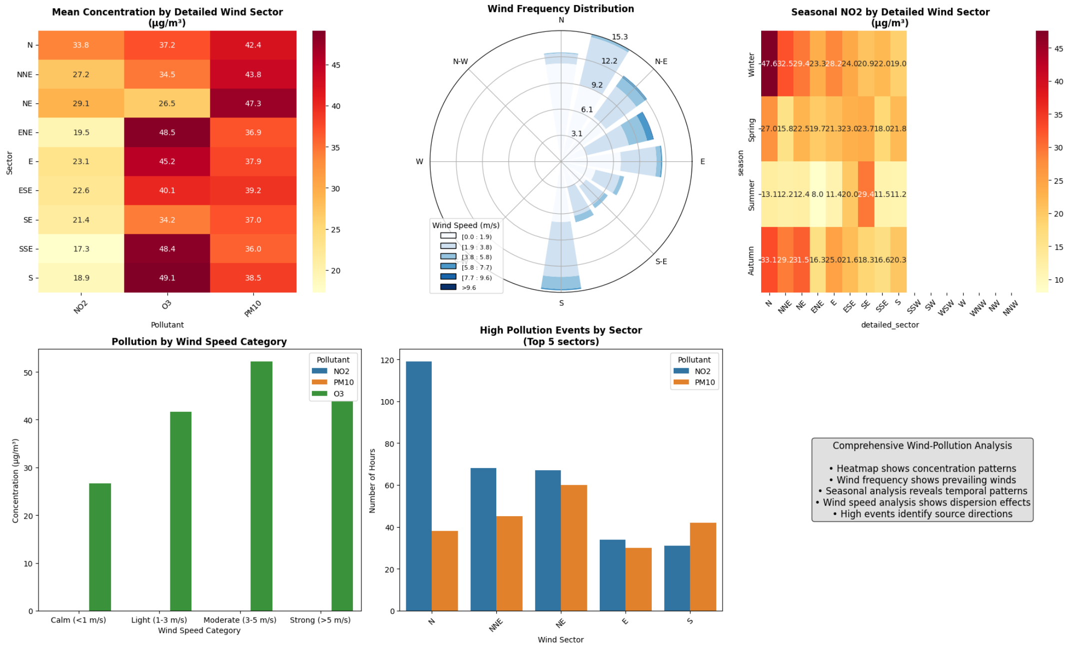


Figure 6: Comprehensive wind-pollution analysis: Mean concentrations by detailed wind sector, wind frequency distribution, seasonal NO profiles, pollution by wind speed category, and high pollution events by sector (Top 5).

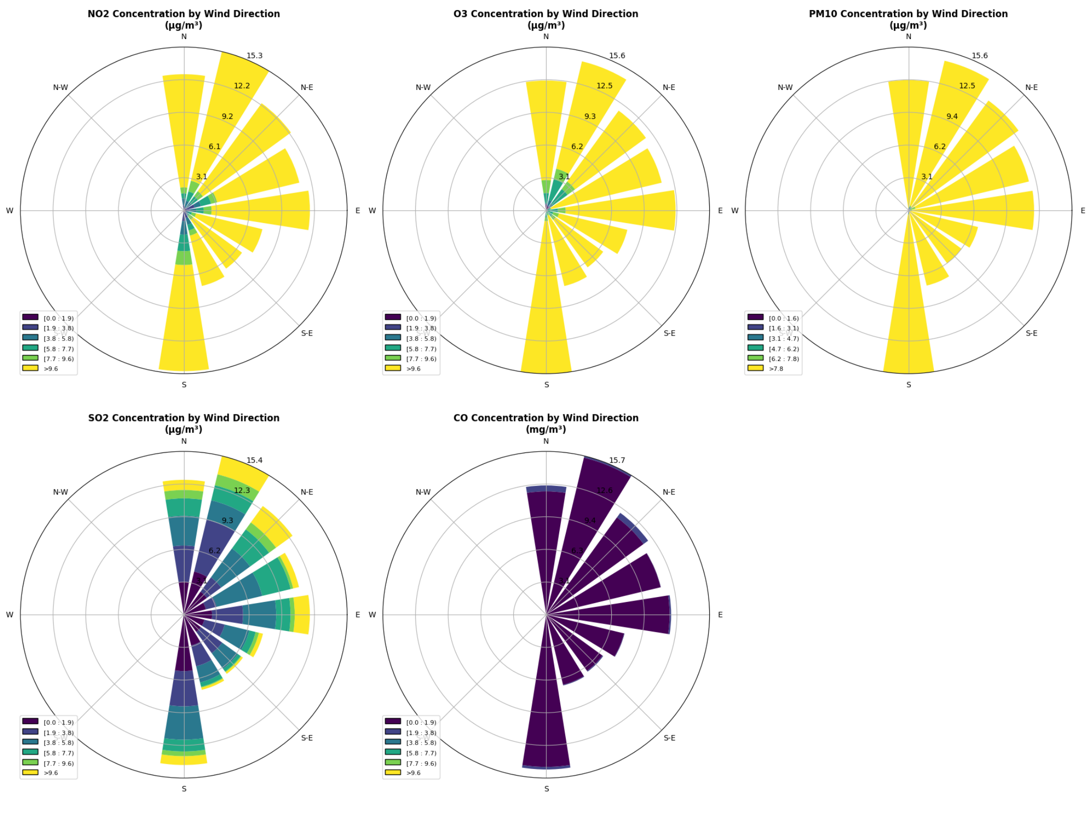


Figure 7: NO2, O3, PM10, SO2, CO concentrations by wind direction.

6 Pollution Episodes

6.1 High Pollution Episodes

Table 10: Sectoral distribution of high pollution episodes

Pollutant	**Hours > Threshold**	**Dominant Sector**	**Distribution**
NO_2 (>50)	388	North (30.7%)	N: 119h, NNE: 68h, NE: 67h
PM_{10} (>75)	287	Northeast (20.9%)	NE: 60h, NNE: 45h, S: 42h
SO_2 (>20)	81	East (25.9%)	E: 21h, NE: 19h, NNE: 11h
CO (>2.0)	69	North (29.0%)	N: 20h, NE: 19h, S: 10h
O_3 (>100)	161	North-Northeast (30.4%)	NNE: 49h, E: 29h, N: 22h

6.2 Analysis of Critical Episodes

Table 11: Characterization of major pollution episodes

Pollutant	**Episode Characteristics**
NO_2	388 hours > 50 g/m^3, mainly from North winds (30.7%), low dispersion conditions
PM_{10}	287 hours > 75 g/m^3, Northeast dominance (20.9%), suggesting industrial and dust sources
SO_2	81 hours > 20 g/m^3, East sector predominant (25.9%), combustion sources identified
CO	69 hours > 2.0 mg/m^3, North sector dominant (29.0%), traffic and incomplete combustion
O_3	161 hours > 100 g/m^3, photochemical formation under NNE winds (30.4%)

The most severe pollution episodes primarily occur:

- Under **North to Northeast winds** for primary pollutants (NO, PM, CO)
- During **calm wind conditions** (<1 m/s) favoring accumulation
- With **stable meteorological configurations** limiting dispersion
- During **periods of high anthropogenic activity** (rush hours, industrial activity)

The recurrence of episodes from specific directions enables the identification of potential source areas and targeting of emission reduction measures.

7 Recommendations

7.1 Corrective Measures

1. **PM_{10}**: Strengthen controls on industrial and construction emissions in the Northeast sector, especially during calm wind periods.
2. **O_3**: Implement an alert system for high photochemical production periods, particularly in summer with high temperatures.
3. **Monitoring**: Increase monitoring frequency during critical seasons (winter for primary pollutants, summer for ozone).
4. **Source identification**: Investigate specific sources in the North and Northeast sectors using dispersion modeling.

7.2 Monitoring System Improvements

- Integrate meteorological forecasting models to anticipate pollution episodes.
- Expand the sensor network for better spatial resolution and source identification.
- Implement a real-time alert system based on observed correlations.

Conclusion

The analysis reveals a concerning situation regarding particulate matter (PM_{10}) and ozone (O_3) at the Hay Hassani station, with exceedances linked to meteorological conditions and local sources. While primary pollutants (NO_2, SO_2, CO) comply with regulatory standards, persistent exceedances of PM_{10} and O_3 require immediate attention and implementation of targeted corrective measures to improve air quality.